Thermo-Electric Conversion of Organic Compounds
using controlled, micro-flow events

It is known that organic molecules can be re-organized using super-critical pressures and temperatures. These changes occur when the molecules are heated to 1200+ degrees F and entrained in a pressure vessel. What is less commonly known is these conditions of pressure and temperature occur inside cavitation bubbles. The principle of this apparatus is the controlled creation of cavitation in such a manner as to entrain a portion of a fluid containing water and organic molecules and subject the fluid to super-critical conditions.

The energy required to affect this change is then that energy required to bring the fluid to sufficient velocity and temperature to cavitate in a controlled manner.

Assume one gallon of fluid being sent through a typical linear venturi:

To induce cavitation requires a velocity of 100 feet per second if the fluid is at 150 degrees F. Using Boyle's law the change in velocity reduces the pressure to less than the vapor pressure of the fluid, inducing cavitation.

The energy required to accelerate the flow to 100 fps is 0.06 hp/gallon or 204.84 btu.
[Eulers Equation: $V^2/2*g = 100^2/2*32.2 = 10000/64.4 = 155.28$ ft= H
hp= Q*H/3960 = (1gal * 155.28)/3960 = 0.039hp
assume efficiency of 65% = 0.039/0.35 = 0.06 hp ]

Each pass through the venturi converts 3% of the fluid.
1hp=2544 BTU
Therefore it requires 204.84/0.03 or 6828 btu.

Based on lab tests, the resulting fluid contains 127,000 btu/ gallon.

Thus the venturi reduces the conversion energy requirement to 5.34% of the energy produced.

How the venturi works:

Cavitation is a phenomenon that occurs under predictable conditions, but location of individual bubbles can not be calculated. The venturi redirects the energy of cavitation, normally associated with a 2 dimensional wave front, first into a controlled tetra-helical vortex, then compresses the three vortices into a rotating flow that also spirals around a center. In this way the energy distribution inside the flow is controlled, directed and non-random.

The primary rotation is nearly 15000rpm. The pressure inside this rotating zone is absolute zero and the temperature is 1500 degrees F. However this zone is only 1000Angstroms in diameter. The flow entrained inside the spiral is rotating at 2000rpm inside a zone 0.04 inches in diameter. Its pressure is also absolute zero and its temperature is 630 degrees F. Since it is not ionized like the flow inside the primary vortex, there is an exchange of energy between the 2 zones causing oscillating zones of gas and liquid to exchange energy in a 'worm-like' motion. In this manner the cavitation is entrained inside

the spiral.

The spiral is formed by the excess energy, commonly called turbulence, being focused to a single location, inducing a local vortex and allowing laminar flow in the rest of the stream. These phenomena are micro-events inside an otherwise stable streamline.